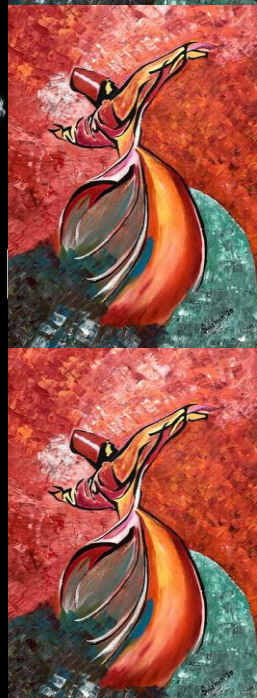




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026

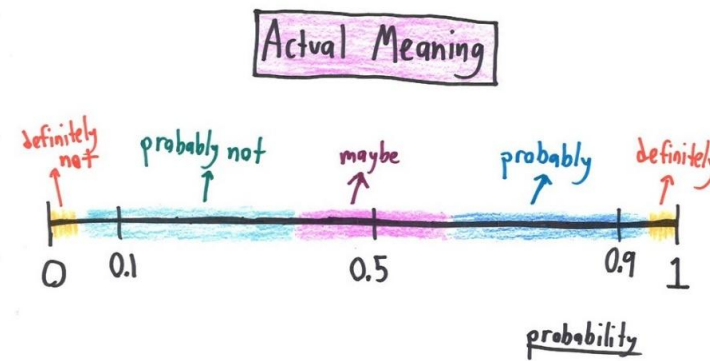




EE 354/CE 361/MATH 310 L4 section (Spring 2026)
Introduction to Probability and Statistics -

Aamir Hasan

Week 3 - Lecture No 06 – January 27, 2026



Announcements!

1. Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)
2. Quiz No 02 (Unit 2) – February 04, 2026
3. Assignment No 01 to be posted by the end of week
4. 03 TAs included in MATH 310 team

Agenda for today

1. Unit 2 – Conditional Probability and Bayes' Rule (to be completed)



Meet Your TAs Rayyan Shah

About Me

I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours

- ❖ Monday 4 – 5 PM
- ❖ Wednesday 4 – 5 PM

Contact Me

✉ rso8770@st.habib.edu.pk

Or “Rayyan Ali Shah” on Teams



Meet Your TAs Hiba Aiman Ali

About Me

I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I'm not in class, I'm most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours

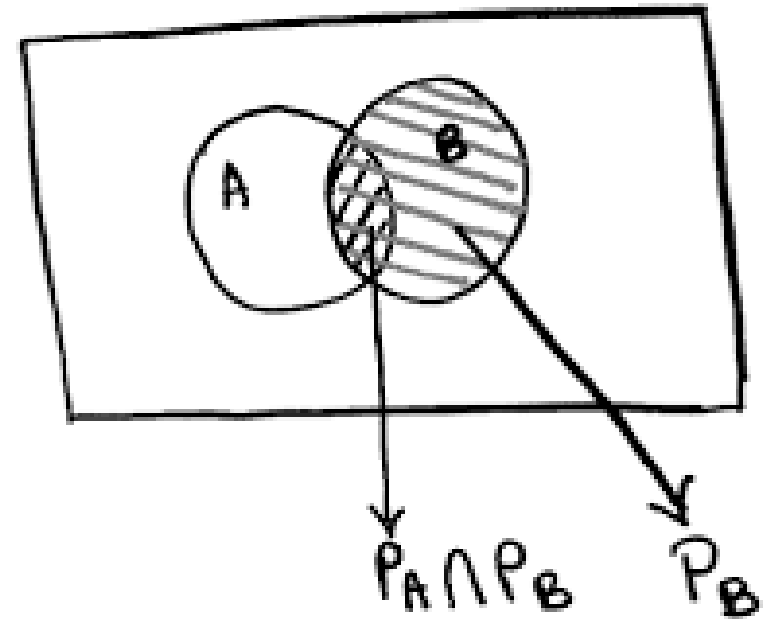
- ❖ Monday 2:30 – 4 PM
- ❖ Wednesday 2:30 – 4 PM
- ❖ Friday 12 – 1 PM

Contact Me

✉ hao9269@st.habib.edu.pk
on Teams or Outlook

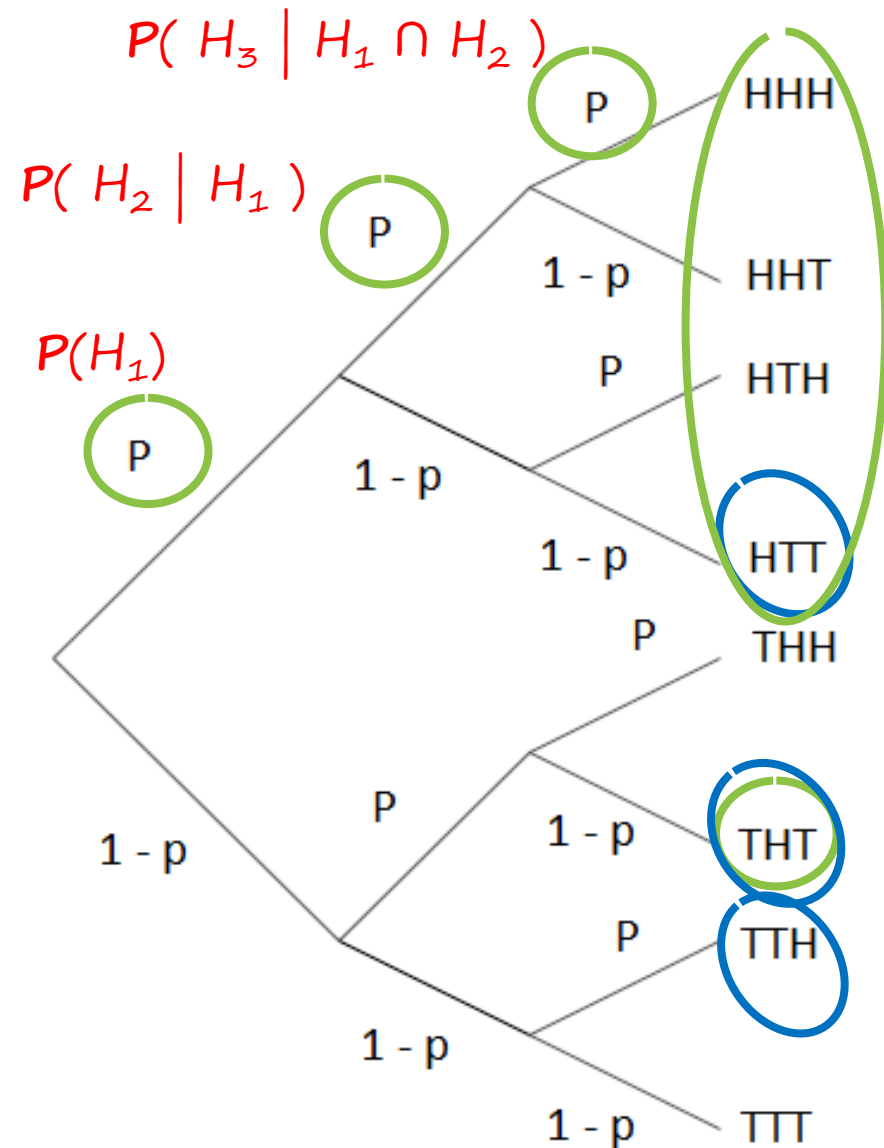
Unit 2: Conditional Probability and Bayes' Rule

- Conditional Probability
- Three important Tools
 - Multiplication rule
 - Total probability theorem
 - Bayes' rule ($\rightarrow$ Inference)



A model based on conditional Probabilities

Let's consider 3 tosses of biased coin: $P(H) = p$; $P(T) = 1 - p$



1. Multiplication Rule: $P(\underline{THT}) = (1-p) p (1-p)$

2. Total Probability: $P(1 \text{ Head}) = 3 p (1-p)^2$

3. Bayes Rule:

$$\begin{aligned} P(\underline{H_1} | 1 \text{ Head}) &= \frac{P(H_1 \cap 1 \text{ head})}{P(1 \text{ head})} \\ &= \frac{p(1-p)^2}{3 p(1-p)^2} = \frac{1}{3} \end{aligned}$$

Few Comments

$$P(H_2 | H_1) = p = P(H_2 | T_1)$$

Any deductions?

$$\begin{aligned} P(H_2) &= P(H_1) P(H_2 | H_1) + P(T_1) P(H_2 | T_1) \\ &= p \cdot p + (1-p) \cdot p \\ &= p \end{aligned}$$

$$\text{So, } P(H_2 | H_1) = p = P(H_2 | T_1)$$

Independence of Two Events

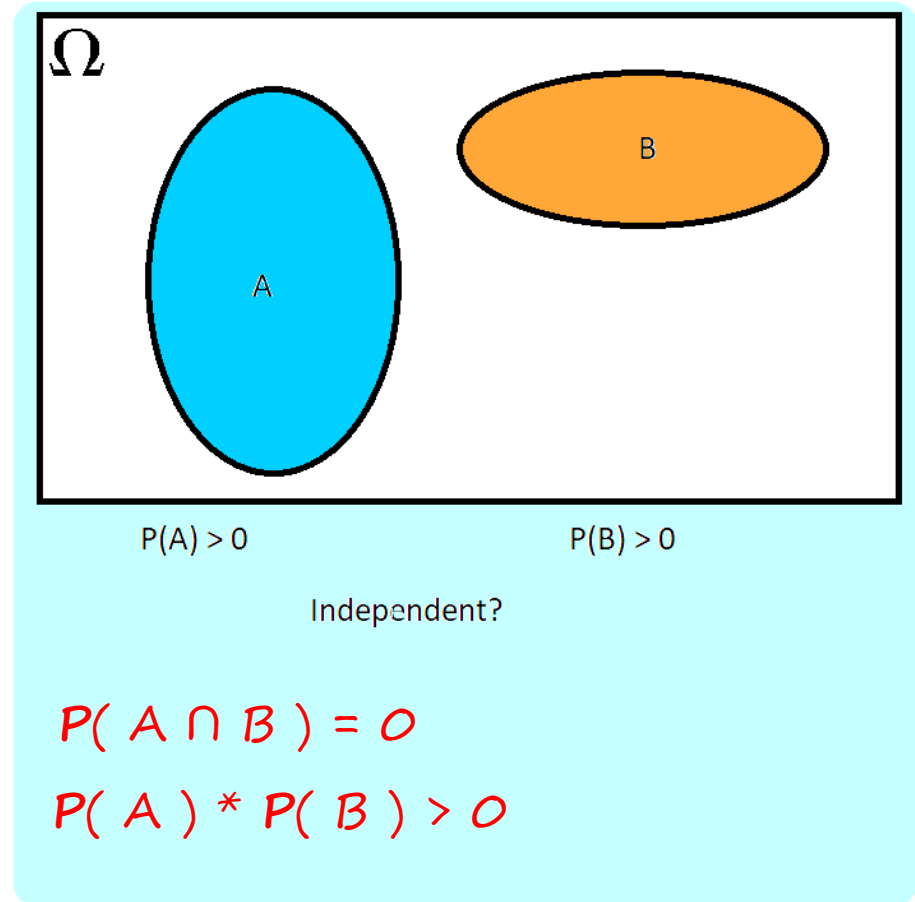
- Intuitive “definition” : $P(B|A) = P(B)$
 - Occurrence of A provides no new information about B

$$\begin{aligned} P(A \cap B) &= P(A) * P(B|A) \\ &= P(A) * P(B) \end{aligned}$$

Definition of Independence: $P(A \cap B) = P(A) * P(B)$

- Symmetric with respect to A and B
- It also implies $P(A|B) = P(A)$
- Also applies even when $P(A) = 0$

Note: Being Independent & being disjoint are different things.



Conclusion?

Two events are independent if occurrence of one tells you nothing about the other.
For example:

A : Flip of a coin

B : It's raining

The above events are physically distinct and non-interacting

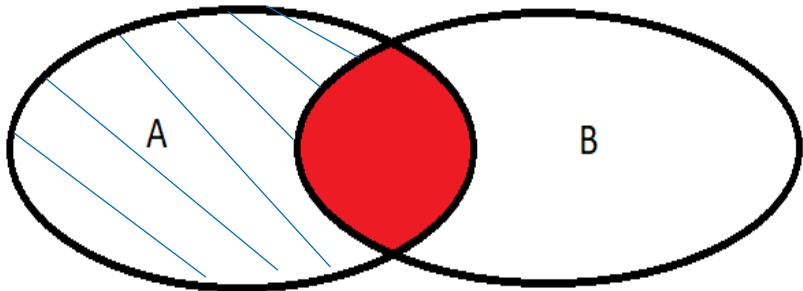
Note: In some cases it is less obvious

Independence of Event Complements

Definition of Independence: $P(A \cap B) = P(A) \cdot P(B)$

- If A and B are independent, then A and B^c are independent.

➤ Intuitive Argument



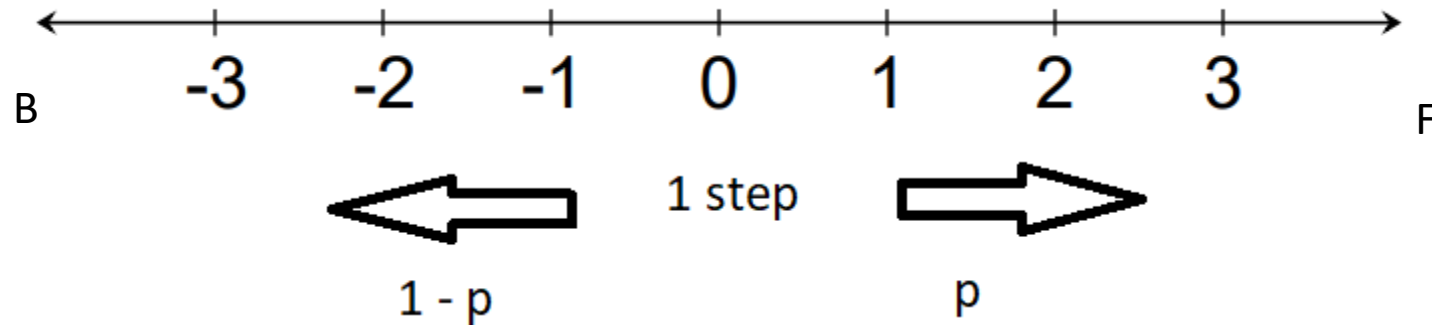
➤ **Formal proof**

$$A = (A \cap B) \cup (A \cap B^c)$$

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B^c) \\ &= P(A)P(B) + P(A \cap B^c) \end{aligned}$$

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A)P(B) \\ &= P(A)(1 - P(B)) \\ &= P(A)P(B^c) \end{aligned}$$

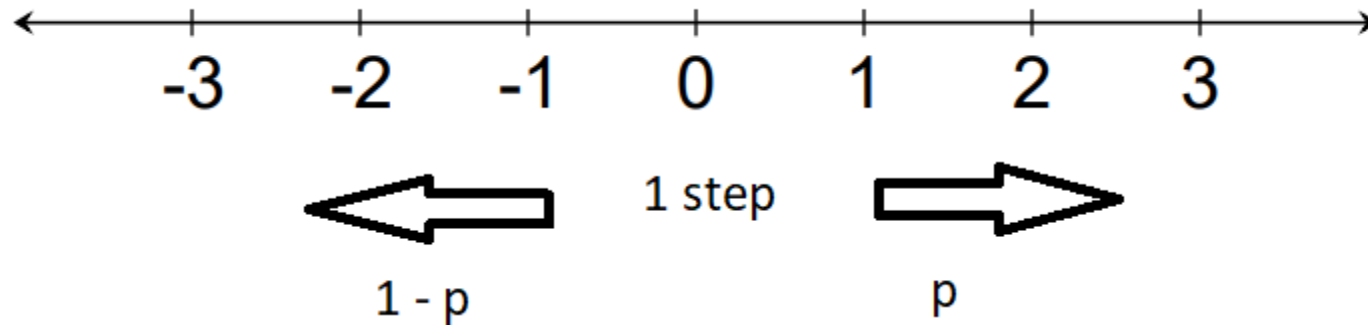
Let's consider an example of a Random Walker



a) $P(\text{After 2 steps the walker is at the origin})$? **Note: Let's call this event A**

$$\begin{aligned} \rightarrow A &= F_1 B_2 \text{ or } B_1 F_2 \\ \rightarrow P(A) &= P(\{ (F_1 B_2), (B_1 F_2) \}) \\ \rightarrow &= P(F_1 B_2) + P(B_1 F_2) \\ \rightarrow &= 2p(1 - p) \end{aligned}$$

Let's consider an example of a Random Walker



b) $P(\text{After 3 steps, the walker is at } +1)$? **Note: Let's call this event B**

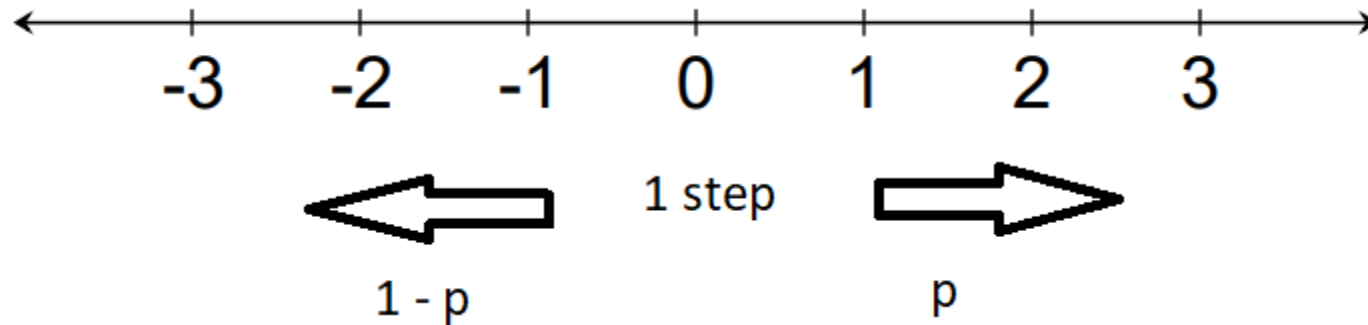
$$\rightarrow B = F_1 F_2 B_3 \text{ or } F_1 B_2 F_3 \text{ or } B_1 F_2 F_3$$

$$\rightarrow P(B) = P(\{ (F_1 F_2 B_3), (F_1 B_2 F_3), (B_1 F_2 F_3) \})$$

$$\rightarrow = 3 p p (1-p)$$

$$\rightarrow = 3 p^2 (1-p)$$

Let's consider an example of a Random Walker



c) $P(1^{\text{st}} \text{ step is backwards} \mid B)$?

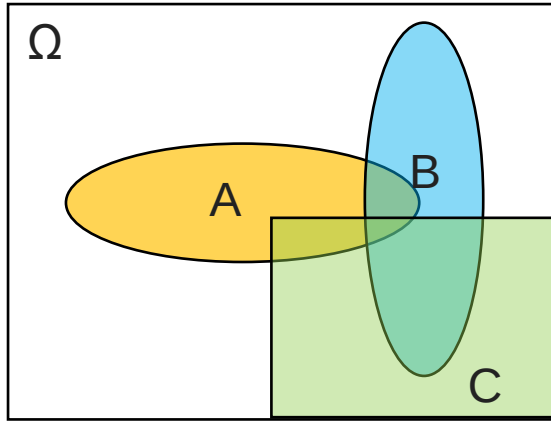
$$B = F_1 F_2 B_3 \text{ or } F_1 B_2 F_3 \text{ or } B_1 F_2 F_3$$

$$P(B) = 3 p^2 (1-p)$$

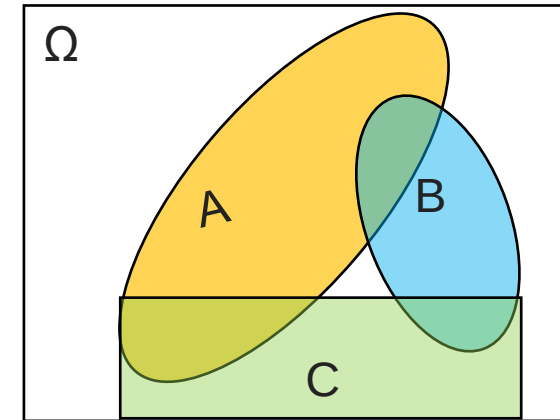
$$\rightarrow \frac{P(B_1 \cap B)}{P(B)} = \frac{P(B_1 F_2 F_3)}{3 p^2 (1-p)} = \frac{p^2 (1-p)}{3 p^2 (1-p)} = \frac{1}{3}$$

Conditional Independence

- Conditional independence, given an event C , is defined as independence under the conditional probability law $P(\cdot \mid C)$



Assume A and B are independent

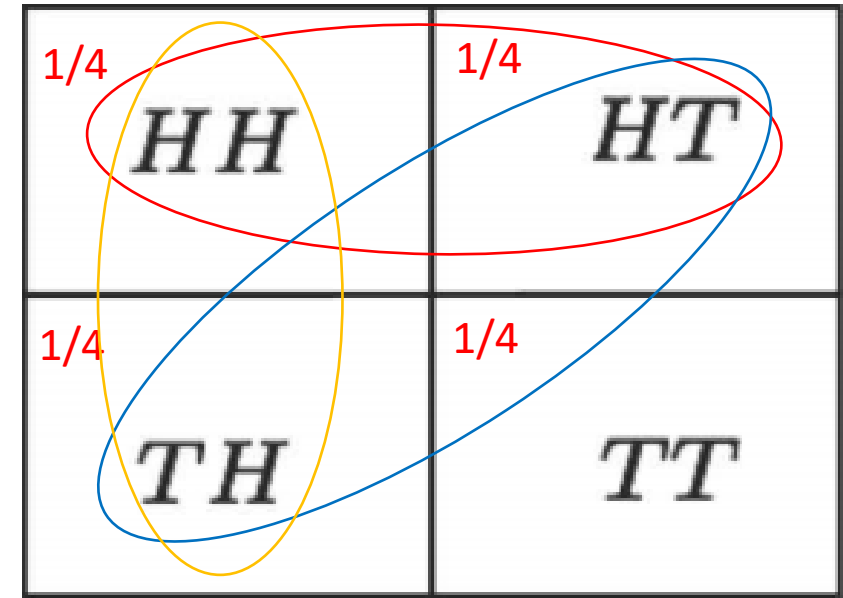


- If we are told that C occurred, are A and B independent? **No**

$$P(A \cap B \mid C) = P(A \mid C) * P(B \mid C)$$

An Example

- Two independent fair coin tosses
 - H_1 : First toss is H
 - H_2 : Second toss is H
 - $P(H_1) = P(H_2) = \frac{1}{2}$
 - C : the two tosses had a different result



H_1 & H_2 are independent events

$(H_1 \cap H_2 \mid C)$ are not independent

$$\begin{aligned} \rightarrow P(H_1 \cap H_2) &= P(HH) = \frac{1}{4} \\ \rightarrow &= P(H_1) \cdot P(H_2) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4} \\ P(H_1 \cap H_2 \mid C) &= 0 \\ \rightarrow &\neq P(H_1 \mid C) \cdot P(H_2 \mid C) = \frac{1}{4} \end{aligned}$$

Puzzle – The King's Sibling

- The king comes from a family of two children. What is the probability that his sibling is female?

Assumptions ➤ Boys have precedence – King is always a boy
➤ $P(\text{Boy}) = P(\text{Girl}) = \frac{1}{2}$, independent

So, What is the answer?

King's sibling has equal chance of being a boy or a girl, therefore, $P(\text{sibling is a girl}) = \frac{1}{2}$ **X**

➔ is this a correct answer? **NO**

➔ Therefore, $P(\text{sibling is a girl}) = \frac{2}{3}$

$B_1 B_2$ $\frac{1}{4}$	$G_1 B_2$ $\frac{1}{4}$
$B_1 G_2$ $\frac{1}{4}$	$G_1 G_2$ $\frac{1}{4}$

➔ It is given that we do have a King

$B_1 B_2$ $\frac{1}{3}$	$G_1 B_2$ $\frac{1}{3}$
$B_1 G_2$ $\frac{1}{3}$	$G_1 G_2$

Independence of a collection of events

- **Intuitively:** Information on some of the events does not change probabilities related to the remaining events.

→ $A_1, A_2, \dots, A_n$ are **independent** $\Rightarrow P(A_3 \cap A_4^c) = P(A_3 \cap A_4^c \mid A_1 \cup (A_2 \cap A_5^c))$

$$\rightarrow P(A_3) = P(A_3 \mid A_1 \cap A_2) = P(A_3 \mid A_1 \cap A_2^c) = P(A_3 \mid A_1^c \cap A_2)$$

Definition: Events $A_1, A_2, \dots, A_n$ are called **independent** if:

$$P(A_i \cap A_j \cap \dots \cap A_m) = P(A_i) P(A_j) \cdots P(A_m) \quad \text{for any distinct indices } i, j, \dots, m$$

$n = 3$:

$$\left. \begin{aligned} P(A_1 \cap A_2) &= P(A_1) \cdot P(A_2) \\ P(A_1 \cap A_3) &= P(A_1) \cdot P(A_3) \\ P(A_2 \cap A_3) &= P(A_2) \cdot P(A_3) \end{aligned} \right\} \text{Pairwise independence}$$

Does pairwise independence **imply** $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$?

Independence vs pairwise independence

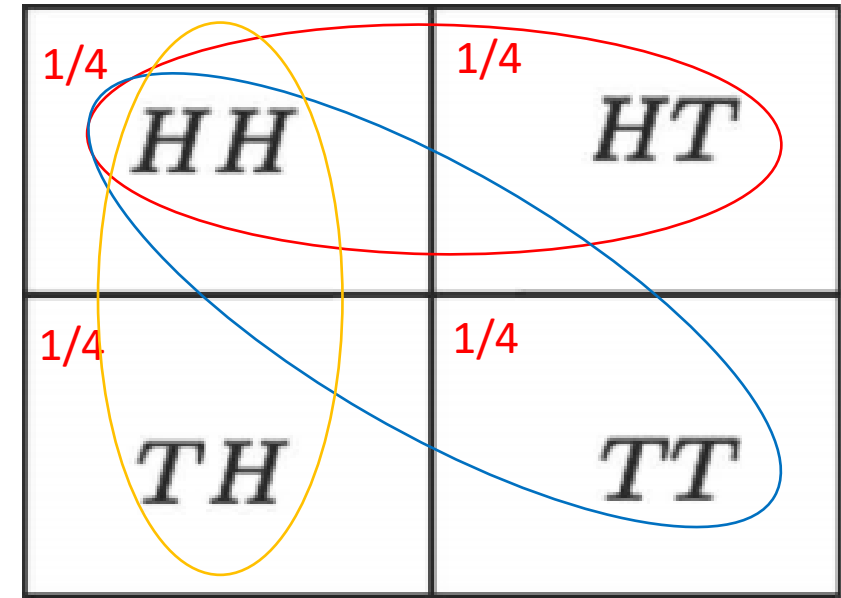
- Two independent fair coin tosses

➤ H_1 : First toss is H

➤ H_2 : Second toss is H

$$P(H_1) = P(H_2) = \frac{1}{2}$$

- C : the two tosses had the same result



$$P(H_1 \cap C) = P(H_1 \cap H_2) = 1/4 \quad P(H_1) * P(C) = \frac{1}{2} \times \frac{1}{2} = 1/4$$

H_1 & C are pairwise independent & so are H_2 & C , & we already know about H_1 & H_2

$$P(H_1 \cap H_2 \cap C) = P(\{HH\}) = 1/4 \quad P(H_1) * P(H_2) * P(C) = 1/8$$

H_1, H_2 & C are collectively not independent

Intuitively, $P(C|H_1) = P(H_2|H_1) = P(H_2) = \frac{1}{2} = P(C)$

$$P(C|H_1 \cap H_2) = 1 \neq P(C)$$

H_1, H_2 and C are pairwise independent, but not independent.

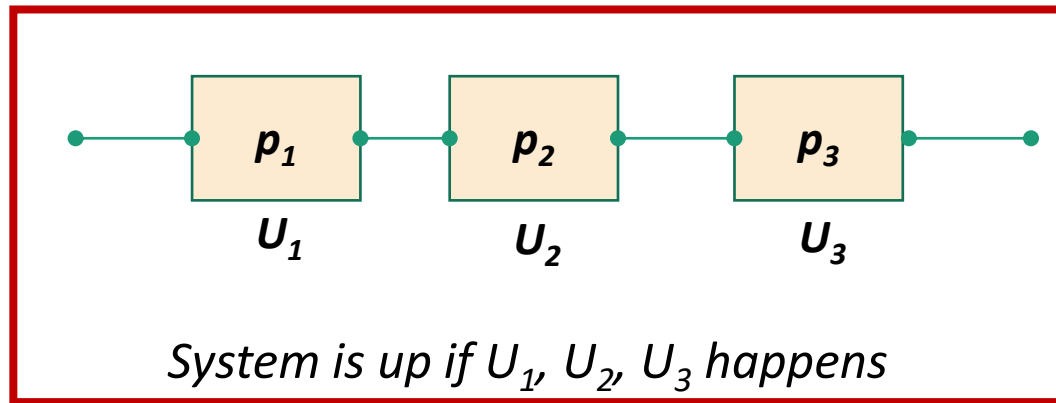
Application of Independence: Reliability (systems in Series)

- p_j : probability that unit j is “up”

Unit i is serviceable w.p p_i & denoted by U_i

Unit i is unserviceable w.p $(1 - p_i)$ & denoted by F_i

➤ $U_{i's}$ are independent, & therefore $F_{i's}$ are also independent



➔
$$\begin{aligned} P(\text{System is up}) &= P(U_1 \cap U_2 \cap U_3) = P(U_1) \cdot P(U_2) \cdot P(U_3) \\ &= p_1 \cdot p_2 \cdot p_3 \end{aligned}$$

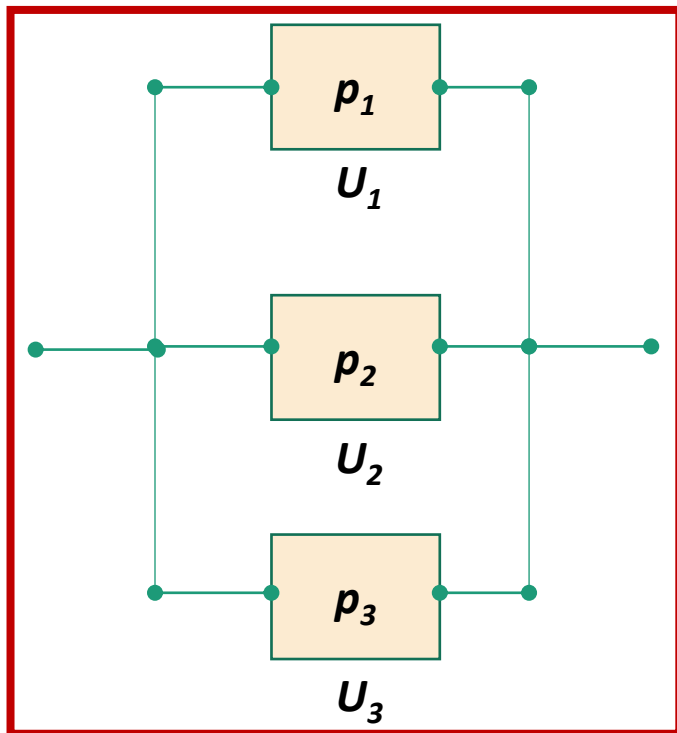
Application of Independence: Reliability (systems in Parallel)

- p_j : probability that unit j is “up”

Unit i is serviceable w.p p_i & denoted by U_i

Unit i is unserviceable w.p $(1 - p_i)$ & denoted by F_i

➤ U_i s are independent, & therefore F_i s are also independent



$$\begin{aligned} \Rightarrow P(\text{System is up}) &= P(U_1 \cup U_2 \cup U_3) \\ \Rightarrow &= 1 - P(U_1 \cup U_2 \cup U_3)^c \\ \Rightarrow &= 1 - P(U_1^c \cap U_2^c \cap U_3^c) \\ \Rightarrow &= 1 - P(F_1 \cap F_2 \cap F_3) \\ \Rightarrow &= 1 - P(F_1) \cdot P(F_2) \cdot P(F_3) \\ \Rightarrow &= 1 - (1 - p_1) \cdot (1 - p_2) \cdot (1 - p_3) \end{aligned}$$

System is up if Path exists between A & B

Practice Problem 1 –

In a specified 6-AM-to-6-AM ($d+1$) 24-hour period, a student wakes up at time t_1 and goes to sleep at some later time t_2 . Assume that 6AM at the start of cycle is 0, 7AM is 1 and so on, so that next day 6AM is time $t = 24$. Therefore, the sample space is given by: –

$$\Omega = \{(t_1, t_2) | t_1 \in [0, 24), t_2 \in (0, 24], \text{ and } t_2 > t_1\}$$

Note – (assume equally likely probability for wake up and sleep time)

Calculate the probability by specifying the set A and sketch the region on the plane corresponding to the event that the “*student is asleep at noon*”.